A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Thermodynamics

This topic is about what heating actually does to matter. You will see where the energy goes when something warms up or melts, why there is a coldest possible temperature, and how the steady pressure of a gas comes from countless tiny collisions. By the end you will be able to derive the link between how fast molecules move and how hot the gas is.

Internal energy and temperature

Everything is made of molecules that are always moving and always pulling or pushing on their neighbours. So every molecule has two kinds of energy: **kinetic energy**, because it moves, and **potential energy**, because of the forces between it and the molecules around it (a bit like the energy stored in a stretched or squashed spring). Add these up over every molecule and you get the **internal energy**.

DEFINITION

Internal energy

The internal energy of a body is the sum of the **randomly distributed** kinetic and potential energies of all of its molecules.

The word **random** matters. At any instant some molecules are fast and some are slow, some are squeezed close to a neighbour and some are far away. You can't say what one molecule is doing, only what the total and the average are.

- **Solid:** molecules vibrate about fixed positions, held in place by strong bonds.
- **Liquid:** molecules are still close together, but they can slide past one another.
- **Gas:** molecules are far apart and move fast in random directions. The forces between them only matter during collisions.

Heating something increases its internal energy, and it can do that in two different ways:

- it can make the molecules move faster on average, raising their **kinetic** energy. You see this as a **rise in temperature**;
- it can pull the molecules further apart against the forces between them, raising their **potential** energy. This is what happens during **melting and boiling**, and the temperature stays **constant** while it happens.

KEY IDEA

Temperature is not internal energy

Temperature is a measure of the **average kinetic energy** of the molecules. Internal energy is the **total** energy of all of them. An iceberg at 0 °C has far more internal energy than a cup of tea at 80 °C, simply because it has vastly more molecules. Energy flows by heating from a higher **temperature** to a lower one, so heat flows from the tea to the iceberg, not the other way.

Specific heat capacity: ΔE = mcΔθ

Think about boiling water in a kettle. With twice as much water you need twice as much energy, because there are twice as many molecules to speed up. To warm the water by 20 K instead of 10 K you also need twice as much energy. So the energy is proportional to both the **mass** and the **temperature change**. The constant of proportionality tells you how much energy a particular material soaks up per kilogram per kelvin, and we call it the specific heat capacity, c.

Heating with no change of state

FORMULA SHEET

$$\Delta E = mc\,\Delta\theta$$

ΔE: energy transferred / J m: mass / kg c: specific heat capacity / J kg⁻¹ K⁻¹
$\Delta\theta$: temperature change / K or °C

DEFINITION

Specific heat capacity

The specific heat capacity of a substance is the energy needed to raise the temperature of **1 kg** of it by **1 K** (or 1 °C), without any change of state.

Water has an unusually large value, $c = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$. Aluminium is about 900 and copper about 390. That is why the sea warms and cools so slowly, and why a metal pan heats up much faster than the water inside it.

EXAM TIP

A temperature **change** of 1 K is exactly the same size as a change of 1 °C, so $\Delta\theta$ can be worked out in either. A temperature itself (like T in $pV = NkT$) must always be in kelvin.

Measuring c with an electric heater

Put an electric heater into a block of the material (or into some water) and measure the voltage V, current I and time t. The energy supplied is $\Delta E = VIt$, so $c = VIt/m\Delta\theta$. If you plot θ against t, the gradient of the straight part is P/mc. Heat lost to the surroundings makes $\Delta\theta$ too small, so c comes out **too large**. Insulating the block, and starting a few degrees below room temperature so that you finish the same amount above it, both reduce this error.

WORKED EXAMPLE

A 2.2 kW kettle contains 0.80 kg of water at 18 °C. (a) Calculate the minimum time it takes to bring the water to 100 °C. (b) In practice it takes 150 s. Calculate the efficiency of the kettle. ($c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$)

Show solution

(a) Energy needed by the water:

$$\Delta E = mc\Delta\theta = 0.80 \times 4180 \times (100 - 18) = 2.74 \times 10^5\ \text{J}$$

The minimum time assumes every joule goes into the water:

$$t = \frac{\Delta E}{P} = \frac{2.742 \times 10^5}{2200} = 125\ \text{s}$$

(b) Energy actually supplied $= 2200 \times 150 = 3.30 \times 10^5$ J, so

$$\text{efficiency} = \frac{2.742 \times 10^5}{3.30 \times 10^5} = 0.83 = 83\%$$

The rest heated the kettle itself and escaped to the room.

Specific latent heat: ΔE = LΔm

Put a thermometer in melting ice and keep heating. The reading sits at 0 °C until the last bit of ice has gone. The energy hasn't disappeared: it is doing work against the forces holding the molecules together, so it increases their **potential** energy instead of their kinetic energy. Because temperature reflects the average kinetic energy, the temperature doesn't change. The word "latent" means hidden, because the thermometer can't see this energy.

The energy needed is proportional to how much substance changes state, which gives another simple equation:

Change of state at constant temperature

FORMULA SHEET

$$\Delta E = L\,\Delta m$$

ΔE: energy transferred / J L: specific latent heat / J kg^{-1} Δm: mass that changes state / kg

DEFINITION

Specific latent heat

The specific latent heat of a substance is the energy needed to change the state of **1 kg** of it **without any change in temperature**. It is called the latent heat of **fusion** for melting (solid ↔ liquid) and of **vaporisation** for boiling (liquid ↔ gas).

For water, $L_\text{f} = 3.34 \times 10^5$ J kg^{-1} and $L_\text{v} = 2.26 \times 10^6$ J kg^{-1}, almost seven times bigger. Melting only has to loosen the structure enough for the molecules to slide past each other. Boiling has to pull every molecule completely away from all of its neighbours, and the escaping vapour also has to push back the atmosphere as it expands.

Part of the heating curve	Equation	What sets its shape (constant power P)
ice warming	$\Delta E = mc_\text{ice}\Delta\theta$	gradient $\dfrac{\Delta\theta}{\Delta t} = \dfrac{P}{mc_\text{ice}}$
melting at 0 °C	$\Delta E = mL_\text{f}$	length $t = \dfrac{mL_\text{f}}{P}$
water warming	$\Delta E = mc_\text{water}\Delta\theta$	gradient $\dfrac{P}{mc_\text{water}}$, half as steep as for ice because $c_\text{water} \approx 2c_\text{ice}$
boiling at 100 °C	$\Delta E = mL_\text{v}$	length $t = \dfrac{mL_\text{v}}{P}$, the longest part by far

COMMON MISTAKE

Don't use $\Delta E = mc\Delta\theta$ across a change of state. Split the process into stages (warm, change state, warm again), find the energy for each stage with the right equation and the right value of c, then add them up.

WORKED EXAMPLE

0.200 kg of ice at −15 °C is heated until it becomes water at 40 °C. (a) Calculate the energy needed. (b) How long does this take with a 500 W heater, assuming no losses? ($c_{\text{ice}} = 2100\ \text{J kg}^{-1}\,\text{K}^{-1}$, $c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$, $L_{\text{f}} = 3.34 \times 10^{5}\ \text{J kg}^{-1}$)

Show solution

Warm the ice from −15 °C to 0 °C:

$$\Delta E_1 = 0.200 \times 2100 \times 15 = 6300\ \text{J}$$

Melt it at 0 °C:

$$\Delta E_2 = 0.200 \times 3.34 \times 10^{5} = 66\,800\ \text{J}$$

Warm the water from 0 °C to 40 °C:

$$\Delta E_3 = 0.200 \times 4180 \times 40 = 33\,440\ \text{J}$$

(a) Total: $\Delta E = 6300 + 66\,800 + 33\,440 = 1.07 \times 10^{5}\ \text{J}$

(b) $t = \dfrac{\Delta E}{P} = \dfrac{1.065 \times 10^{5}}{500} = 213\ \text{s}$. Notice that melting takes more energy than the other two stages put together.

Core Practical 13: specific latent heat

CORE PRACTICAL

Determine the specific latent heat of vaporisation of water

Stand a beaker of water on a top-pan balance and put an immersion heater in it, connected through a joulemeter (or an ammeter, a voltmeter and a stopwatch, so that $E = VIt$). Bring the water to the boil and let it boil steadily. Then record the balance reading, start the clock, and record the reading again after a measured time (about 5 minutes).

The energy supplied boils off a mass Δm, so $L = \dfrac{E}{\Delta m} = \dfrac{Pt}{\Delta m}$.

A single run always gives too large a value, because some of the energy heats the room instead of boiling water. The better method is to repeat at several different powers and plot **power P against the rate of mass loss** $\Delta m/\Delta t$. If heat is lost at a steady rate P_{loss}:

$$P = L\,\frac{\Delta m}{\Delta t} + P_{\text{loss}}$$

This is $y = mx + c$: the **gradient is** L and the intercept on the power axis is the rate of heat loss, which the gradient method removes.

- Only start timing once the water is boiling steadily, otherwise some energy is still raising its temperature.
- Insulate the sides of the beaker to cut heat loss. Keep the heater fully covered by water.
- Splashing makes Δm too big, and steam condensing on the heater and dripping back makes it too small.
- **Safety:** steam and boiling water scald. Switch the heater on only when it is immersed, keep the connections dry, and let everything cool before you move it.

For the latent heat of **fusion** you can use a heater in a funnel of crushed ice and collect the meltwater. A second, identical funnel with no heater shows how much ice melts just from the warmth of the room, and you subtract that mass.

Core Practical 12: a thermistor thermostat

A **negative temperature coefficient (NTC) thermistor** is a semiconductor whose resistance **falls** as it warms up, because extra thermal energy frees more charge carriers. Put it in series with a fixed resistor R across a supply V_{in} and you have a potential divider. The p.d. across the fixed resistor is

$$V_{\text{out}} = V_{\text{in}} \frac{R}{R + R_{\text{th}}}$$

As the temperature rises, R_{th} falls, so the fixed resistor takes a bigger share of the supply and V_{out} **rises**. Connect V_{out} to a switching circuit (a transistor or relay) that turns a heater off above a set voltage, and you have a **thermostat**. To set it to a particular temperature you first need to know which voltage goes with which temperature. Finding that out is called **calibration**.

CORE PRACTICAL

Calibrate a thermistor in a potential divider circuit as a thermostat

Put the thermistor (waterproofed, or inside a test tube) in a beaker of water next to a thermometer. Start with ice and water near 0 °C and warm it slowly. At each temperature, stir, wait until the readings are steady, and record θ and V_{out}. Plot a **calibration curve** of V_{out} against θ.

To use the circuit as a thermostat, read the voltage that goes with your target temperature off the curve and set that as the switching voltage. The circuit is most sensitive (biggest change in voltage per kelvin) where the curve is steepest, which is around the temperature at which $R_{\text{th}} = R$. So choose a fixed resistor about equal to the thermistor's resistance at the temperature you care about.

- The thermistor and thermometer must be at the same temperature: stir, keep them close together and wait before each reading.
- Keep the current small so the thermistor doesn't heat itself.
- Read the thermometer at eye level to avoid parallax.

- **Safety:** use a low-voltage supply near water, and take care with hot water.

Absolute zero and the kelvin scale

Seal some air in a flask so its volume can't change, and heat it. The pressure rises in a straight line with temperature. Now imagine cooling it instead and follow the line down. Whatever gas you use, and however much of it, the line reaches **zero pressure at about −273 °C**.

Zero pressure means the molecules have stopped hitting the walls: they have the least kinetic energy they can possibly have. You can't cool anything below this, because you can't take away kinetic energy that isn't there. This temperature is called **absolute zero**, and the kelvin scale starts from it.

DEFINITION

Absolute zero

Absolute zero (0 K, which is −273 °C) is the lowest possible temperature. It is the temperature at which the molecules of a substance have their **minimum possible kinetic energy**, so the substance has its minimum internal energy.

$$T/\mathrm{K} = \theta/^{\circ}\mathrm{C} + 273$$

Because the kelvin scale starts at absolute zero, the **average kinetic energy of the molecules is directly proportional to the absolute temperature** T. Twice the kelvin temperature means twice the average kinetic energy. Celsius doesn't work like that: going from 20 °C to 40 °C is going from 293 K to 313 K, which increases the average kinetic energy by only about 7%.

COMMON MISTAKE

Always convert to kelvin before using any gas equation. In Celsius, warming a gas from 10 °C to 20 °C looks as if it should double the pressure. In kelvin it is 283 K → 293 K, a rise of only 3.5%.

The ideal gas equation: pV = NkT

Experiments on a fixed amount of gas give three simple rules:

- **Boyle's law:** at constant temperature, $p \propto 1/V$. Squash the gas into half the space and the molecules hit each bit of wall twice as often.
- **Pressure law:** at constant volume, $p \propto T$. Hotter molecules hit the walls harder and more often.
- **Charles's law:** at constant pressure, $V \propto T$.

Pressure is also proportional to the number of molecules N: double the gas in the same box, double the collisions. Put all of these together and you get one equation, where the constant k is the **Boltzmann constant**, $1.38 \times 10^{-23}\ \text{J K}^{-1}$.

Ideal gas equation

FORMULA SHEET

$$pV = NkT$$

p: pressure / Pa V: volume / m³ N: number of molecules k: Boltzmann constant / J K⁻¹
T: absolute temperature / K

An **ideal gas** is one that obeys this equation exactly. Real gases come very close as long as the pressure isn't too high and the temperature is well above the point where the gas would turn into a liquid. The model behind it (the kinetic theory, below) assumes:

- a very large number of molecules moving in random directions with a range of speeds;
- the molecules themselves take up a negligible fraction of the container's volume;
- there are no forces between molecules except during collisions;
- all collisions are elastic, and each lasts a negligible time compared with the time between collisions.

EXAM TIP

For a fixed amount of gas N and k are constant, so $\frac{pV}{T}$ is constant: $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$. Any units of p and V work in this ratio form, as long as they're the same on both sides, but T must be in kelvin.

WORKED EXAMPLE

A helium cylinder of volume 0.050 m³ holds gas at 1.5 × 10⁷ Pa and 20 °C. (a) How many helium atoms does it contain? (b) What is the mass of the helium? (mass of a helium atom = 4.0 u) (c) How many balloons of volume 0.010 m³ at 1.0 × 10⁵ Pa can be filled, if the temperature stays at 20 °C?

Show solution

(a) $T = 20 + 273 = 293\ \text{K}$, so

$$N = \frac{pV}{kT} = \frac{1.5 \times 10^7 \times 0.050}{1.38 \times 10^{-23} \times 293} = 1.85 \times 10^{26}$$

(b) $m = 1.85 \times 10^{26} \times 4.0 \times 1.66 \times 10^{-27} = 1.23\ \text{kg}$

(c) At constant temperature $p_1V_1 = p_2V_2$, so at balloon pressure the gas would take up

$$V_2 = \frac{1.5 \times 10^7 \times 0.050}{1.0 \times 10^5} = 7.5 \text{ m}^3$$

But 0.050 m³ of that has to stay behind in the cylinder (it can't go below balloon pressure), leaving 7.45 m³ for balloons: $7.45 / 0.010 = 745$ balloons.

Core Practical 14: pressure and volume

CORE PRACTICAL

Investigate the relationship between pressure and volume of a gas at fixed temperature

A common set-up is the Boyle's law apparatus: air is trapped in a sealed glass tube above a column of oil. A foot pump pushes more oil in, and a Bourdon gauge reads the pressure. The tube has a constant cross-section, so the length of the air column is proportional to its volume. (A syringe connected to a pressure sensor works the same way.)

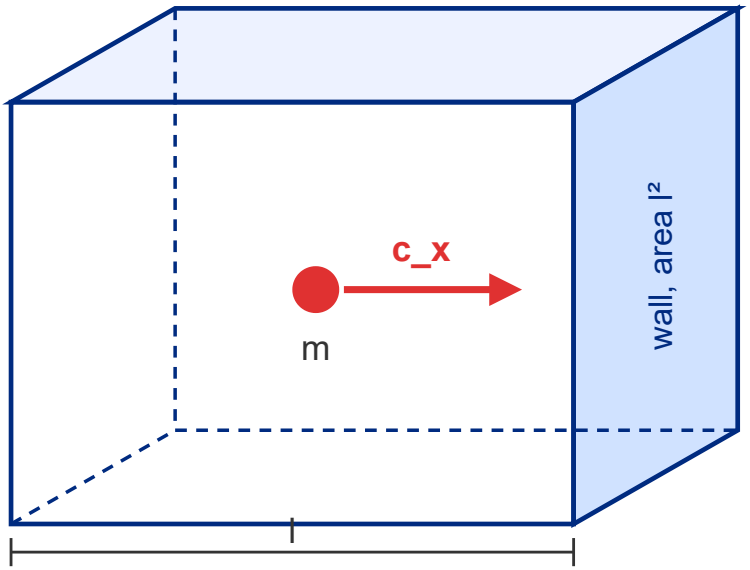
Change the pressure in steps. After each change, **wait** for a minute: compressing a gas warms it, and the law only holds at constant temperature. Then record p and V.

A graph of p against V is a curve, which is hard to test. Plot p **against** $1/V$ instead. From $pV = NkT$, $p = NkT \times \frac{1}{V}$, so the graph should be a **straight line through the origin with gradient** NkT.

- Read the oil level at eye level to avoid parallax.
- Check for leaks: the readings should return to their starting values when you release the pressure.
- Gas in the tubing and the gauge that isn't counted in your volume reading (“dead volume”) is a systematic error. It makes every true volume bigger than the reading.
- **Safety:** the glass tube is under pressure, so wear eye protection or use a safety screen, and never exceed the gauge's maximum pressure.

Kinetic theory: where pressure comes from

Why does a gas push on its container at all? Molecules keep hitting the walls and bouncing off. Every bounce reverses a molecule's momentum, and by Newton's second law a change of momentum needs a force. Countless collisions every second add up to a steady force on each part of the wall, and force per unit area is pressure. Let's turn that picture into an equation.



before: +mc_x

after: −mc_x

Δp = −2mc_x

back again after
Δt = 2l / c_x

One molecule in a cube of side l. Only the velocity component c_x towards the shaded wall matters for collisions with that wall.

Deriving pV = ⅓Nm⟨c²⟩ from molecules hitting the walls

1 A molecule of mass m moves towards the right-hand wall with velocity component c_x. The collision is elastic, so it bounces back with $-c_x$. Its change of momentum is

$$\Delta p = (-mc_x) - (mc_x) = -2mc_x$$

By Newton's third law the wall receives an equal and opposite momentum, $+2mc_x$.

2 Before it hits that wall again it has to cross to the far wall and come back, a distance $2l$, at speed c_x:

$$\Delta t = \frac{2l}{c_x}$$

Bounces off the other walls don't change c_x, so they don't affect this time.

3 The average force this one molecule exerts on the wall is the rate at which it delivers momentum:

$$F = \frac{\Delta p}{\Delta t} = \frac{2mc_x}{2l/c_x} = \frac{mc_x^2}{l}$$

4 There are N molecules, each with its own c_x. Add up their forces and write the sum using the mean value of c_x^2:

$$F = \frac{m}{l}\left(c_{x1}^2 + c_{x2}^2 + \cdots + c_{xN}^2\right) = \frac{Nm\langle c_x^2\rangle}{l}$$

5 Pressure is force divided by area, and the wall's area is l^2. Also $l^3 = V$, the volume of the box:

$$p = \frac{F}{l^2} = \frac{Nm\langle c_x^2\rangle}{l^3} = \frac{Nm\langle c_x^2\rangle}{V}$$

6 Each molecule's speed c satisfies $c^2 = c_x^2 + c_y^2 + c_z^2$ (Pythagoras in 3D). The motion is random, so no direction is special and on average the three components are equal:

$$\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \tfrac{1}{3}\langle c^2\rangle$$

7 Substitute and multiply both sides by V:

$$p = \frac{Nm\langle c^2\rangle}{3V} \quad \Rightarrow \quad \boxed{pV = \tfrac{1}{3}Nm\langle c^2\rangle}$$

Collisions between molecules don't spoil this: they only swap momentum between molecules, and the random spread of velocities stays the same on average. The cube shape doesn't matter either.

$pV = \frac{1}{3}Nm\langle c^2\rangle$ is **not on the formula sheet**, but you need it: it is the starting point of the derivation that the spec does require, which comes next.

KEY IDEA

What ⟨c²⟩ means

$\langle c^2\rangle$ is the **mean square speed**: square every molecule's speed, then take the average. It is not the same as the mean speed squared. For three molecules moving at 1, 2 and 3 m s⁻¹, the mean speed is 2 m s⁻¹, but $\langle c^2\rangle = (1+4+9)/3 = 4.67\ \text{m}^2\,\text{s}^{-2}$. Its square root is the **root mean square (r.m.s.) speed**, $c_{\text{rms}} = \sqrt{\langle c^2\rangle} = 2.16\ \text{m s}^{-1}$.

Linking molecular speed to temperature

Now we have two equations for pV: one from experiments ($pV = NkT$) and one from a model of bouncing molecules ($pV = \frac{1}{3}Nm\langle c^2\rangle$). Both describe the same gas, so they must agree.

Deriving ½m⟨c²⟩ = 3/2 kT

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Write down the two expressions for pV: kinetic theory, and the ideal gas equation.

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle \qquad \text{and} \qquad pV = NkT$$

2 They describe the same gas, so set them equal:

$$\tfrac{1}{3}Nm\langle c^2\rangle = NkT$$

3 N appears on both sides, so cancel it, then multiply both sides by 3:

$$m\langle c^2\rangle = 3kT$$

4 Divide both sides by 2, so the left-hand side becomes the average kinetic energy of one molecule:

$$\boxed{\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT}$$

$\frac{1}{2}m\langle c^2\rangle$ is the mean of $\frac{1}{2}mc^2$ over all molecules: the average (translational) kinetic energy per molecule.

Average kinetic energy of a gas molecule

FORMULA SHEET

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

m: mass of one molecule / kg $\langle c^2\rangle$: mean square speed / m² s⁻² k: Boltzmann constant / J K⁻¹
T: absolute temperature / K

This is the result behind the earlier statements. The average kinetic energy of a molecule depends **only on the absolute temperature**: not on the pressure, not on the volume, and not on which gas it is. At the same temperature a heavy xenon atom and a light helium atom have the same average kinetic energy, so the helium atom must be moving much faster. At $T = 0$ the average kinetic energy would be zero, which is the idea of absolute zero again.

The r.m.s. speed of the molecules

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Start from $m\langle c^2\rangle = 3kT$ and divide by m:

$$\langle c^2\rangle = \frac{3kT}{m}$$

2 Take the square root to get the r.m.s. speed:

$$\boxed{c_{\text{rms}} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}}}$$

So $c_{\text{rms}} \propto \sqrt{T}$: to double the speed of the molecules you need four times the kelvin temperature.

WORKED EXAMPLE

Air is mostly nitrogen molecules, each of mass 28 u. For air at 20 °C, calculate (a) the average kinetic energy of a nitrogen molecule, (b) the r.m.s. speed of the molecules, and (c) the temperature at which the r.m.s. speed would be twice as big.

Show solution

(a) $T = 293$ K:

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21}\ \text{J}$$

(b) $m = 28 \times 1.66 \times 10^{-27} = 4.65 \times 10^{-26}$ kg, so

$$c_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 293}{4.65 \times 10^{-26}}} = 511\ \text{m s}^{-1}$$

That is faster than the speed of sound in air (about 340 m s⁻¹), which makes sense: sound is carried by these molecules.

(c) $c_{\text{rms}} \propto \sqrt{T}$, so doubling the speed needs $4T = 4 \times 293 = 1172$ K, which is about 900 °C.

Quick checks

QUICK CHECK

Ice at 0 °C melts into water at 0 °C. What happens to the molecules?

- Their average kinetic energy increases and their potential energy stays the same.
- Their potential energy increases and their average kinetic energy stays the same.
- Both their kinetic and potential energies stay the same, because the temperature doesn't change.
- Their internal energy decreases as the bonds break.

The temperature is constant, so the average kinetic energy is constant. Energy is still being supplied, so the internal energy rises: it goes into potential energy as the molecules are pulled apart against the forces between them.

QUICK CHECK

A heater of constant power warms some ice and then the water it melts into. Why is the θ–t graph steeper for the ice than for the water?

- Ice has a lower density than water.
- Some of the energy is used to melt the ice.
- Ice has a smaller specific heat capacity than water.
- Water loses more heat to the surroundings.

The gradient is $\Delta\theta/\Delta t = P/mc$. The mass and power are the same for both, and $c_{\text{ice}} \approx 2100$ is about half of $c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$, so the ice warms about twice as fast.

QUICK CHECK

A gas is heated from 27 °C to 327 °C. By what factor does the r.m.s. speed of its molecules increase?

- 12.1
- 2.0
- 1.41
- 1.10

Convert to kelvin: 300 K → 600 K, so T doubles. $c_{\text{rms}} = \sqrt{3kT/m} \propto \sqrt{T}$, so the speed goes up by $\sqrt{2} = 1.41$. Using Celsius (327/27 = 12.1) is the classic trap.

QUICK CHECK

Helium and xenon are mixed in a container and left to reach the same temperature. Which quantity is the same for both kinds of atom?

- Their r.m.s. speed
- Their momentum
- Their average kinetic energy
- The number of times per second they hit the walls

$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ depends only on T. Xenon atoms are about 33 times more massive, so they move $\sqrt{33} \approx 5.7$ times more slowly, and they hit the walls less often.

QUICK CHECK

A fixed mass of ideal gas has its volume doubled while its kelvin temperature is tripled. What happens to its pressure?

- It falls to 2/3 of its original value.
- It increases to 1.5 times its original value.
- It increases to 6 times its original value.
- It stays the same.

$p = NkT/V$, so $p \propto T/V$. Tripling T and doubling V gives a factor of $3/2 = 1.5$.

QUICK CHECK

An NTC thermistor and a fixed resistor form a potential divider. The output is the p.d. across the fixed resistor. As the temperature rises, what happens to the output?

- It falls, because the thermistor's resistance rises.
- It falls, because the thermistor's resistance falls.
- It rises, because the thermistor's resistance falls.
- It stays constant, because the supply p.d. is constant.

For an NTC thermistor, resistance falls as temperature rises. The fixed resistor then has a bigger share of the total resistance, so it takes a bigger share of the supply p.d.: $V_{\text{out}} = V_{\text{in}}R/(R + R_{\text{th}})$ rises.